

? Salesforce Summer Program – Today’s Target (4–5 Hours)

☐ Goal for Today

By end of today, you should understand:

- Why testing matters in enterprise systems
 - What is Asynchronous Apex
 - Why developers use Salesforce DX and CLI
 - How professional Salesforce development workflow operates
 - How all concepts learned so far fit into one complete system
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☐ Deep Learning Modules (IMPORTANT)

1 ☐ Apex Testing (~2 hrs 15 mins)

Focus:

- Why testing matters
- Unit tests
- Reliability
- Preventing bugs

☐ Do NOT focus on memorizing syntax.

2 ☐ Quick Start: Salesforce DX (~40 mins)

Focus:

- Salesforce DX workflow
 - Source-driven development
 - GitHub integration
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☐ Light Completion Modules

3 ☐ Asynchronous Apex (~overview only)

Focus:

- Why async processing exists
- Future methods
- Queueable Apex

☐ High-level understanding is enough for now.

4 ☐ Command-Line Interface (Quick Revision)

Focus:

- CLI workflow
 - Productivity
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☐ Videos (Verified Working Links)

1 ☐ Apex Testing Basics

https://www.youtube.com/watch?v=nMyKE6_U5qo

Focus:

- Why testing matters
 - Apex test classes
 - Enterprise reliability
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2 ☐ What is Apex Testing?

<https://www.youtube.com/watch?v=RK7uSw1ynLI>

Focus:

- Unit testing basics
 - Why Salesforce requires testing
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3 ☐ Asynchronous Apex Explained

<https://www.youtube.com/watch?v=Lex6jrtB0Ho>

Focus:

- Async processing
 - Background jobs
 - Why systems need async execution
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4 Introduction to Salesforce DX

<https://www.youtube.com/watch?v=HV9fZei0-xs>

Focus:

- Salesforce DX workflow
 - Modern developer experience
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5 Salesforce CLI & DX Workflow

<https://www.youtube.com/watch?v=woINCxIT4ng>

Focus:

- Salesforce CLI
 - VS Code integration
 - Development workflow
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CORE TASKS (MANDATORY)

Task 1 System Integration Exercise (MOST IMPORTANT)

Take your College Management System.

Now explain COMPLETE workflow from beginning to end:

Example Flow:

Student registers →
Validation Rules check data →
Flow sends confirmation →
Trigger updates course count →
Formula recalculates seats →
Platform Event sends notification →
Database stores records →
Reports show analytics

☐ Explain each step properly.

Task 2 Testing Thinking

Suggest 5 things that MUST be tested in your system.

Examples:

- Invalid email
- Duplicate registration
- Course overbooking
- Attendance calculation
- Trigger execution

Explain:

☐ What problem could happen if not tested?

Task 3 Async Thinking

Give 3 examples where:

☐ background processing is better than immediate processing.

Examples:

- Sending bulk emails
 - Large report generation
 - Data synchronization
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Task 4 Developer Workflow Reflection

Answer:

☐ Why do professional developers use:

- GitHub
 - DX
 - CLI
- instead of only browser clicks?
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☐ **GitHub Submission**

Create:

/day7-testing-dx/

README.md must include:

1. Why Testing Matters

2. What is Asynchronous Apex?

3. What is Salesforce DX?

4. Complete System Workflow

(End-to-end explanation)

5. Important Test Cases

(Your examples)

6. Reflection

Why enterprise software development needs structured workflows

☐ Revision Questions

1. Why are tests important in enterprise systems?
 2. What problems happen without testing?
 3. Why is asynchronous processing useful?
 4. Difference between synchronous and asynchronous processing?
 5. Why do developers use version control?
 6. Why is GitHub important?
 7. Why is DX useful for teams?
 8. How do Flows, Triggers and Validation Rules work together?
 9. Why should business logic be tested carefully?
 10. Why is developer workflow important in large teams?
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☐ ☐ Rules

- Focus on workflow understanding
 - Think like a professional developer
 - Use real-world examples
 - Do NOT memorize syntax blindly
 - Spend around 4–5 hours maximum
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☐ End of Today Outcome

You should now clearly understand:

- ☐ Why testing matters
 - ☐ Why async processing exists
 - ☐ Professional Salesforce development workflow
 - ☐ Importance of GitHub + DX + CLI
 - ☐ How all Salesforce concepts integrate into one system
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